82-0722

# AEROELASTIC FLUTTER AND DIVERGENCE OF STIFFNESS COUPLED, GRAPHITE/EPOXY, CANTILEVERED PLATES

A82-30172

Steven J. Hollowell*
Flight Dynamics Laboratory
Air Force Wright Aeronautical Laboratories
Wright Patterson Air Force Base, OH 45433

John Dugundji**
Department of Aeronautics and Astronautics
Massachusetts Institute of Technology
Cambridge, MA 02139



## Abstract

An analytical and experimental investigation was conducted to determine the aeroelastic flutter and divergence behavior of unswept, rectangular wings simulated by graphite/epoxy, cantilevered plates with varying amounts of bending-torsion stiffness coupling. The analytical approach incorporated a Rayleigh-Ritz energy formulation and unsteady, incompressible 2-dimensional aerodynamic theory. Flutter and divergence velocities were obtained using the V-g method and compared to results of low-speed wind tunnel tests. Stall flutter behavior was also examined experimentally. There was good agreement between analytical and experimental results. Wings with negative stiffness coupling exhibited divergence, while positive coupling delayed the onset of stall flutter.

## 1. Introduction

Aeroelastic tailoring is broadly defined as the technology to design a lifting surface which exhibits a desired aeroelastic response. Desired aeroelastic responses most often considered are maximization of flutter and divergence speeds, although other aeroelastic responses can be important depending on the application, such as control reversal speed, camber changes as a function of load and speed and angle of attack changes as a function of load.

Aeroelastic tailoring, which exploits the anisotropic character of advanced composites, has received considerable attention in recent literature. N. J. Krone, Jr.[1] concluded that forward swept wings without divergence or weight penalties may be possible through the use of selectively laminated advanced composites. T. A. Weisshaar[2,3] extended, analytically, Krone's conclusion to potentially practical wing designs. Weisshaar concluded that the bending-torsion stiffness coupling of anisotropic advanced composite materials could be used to alleviate divergence in forward swept wings. V. C. Sherrer, T. J. Hertz, and M. H. Shirk[4] conducted a series of wind tunnel tests using simple plate-like models of a forward swept wing. These tests essentially verified Weisshaar's conclusion, and also showed that existing analytical techniques (computer programs) would adequately predict the divergence dynamic pressures for most test conditions.

Additional aeroelastic wind tunnel tests on specific forward swept wing model designs were conducted by Wilkinson and Rauch[5] of Grumman Aircraft and Ellis, Dobbs, and Miller[6] of Rockwell North American Aircraft. A good general discussion of these efforts on forward swept wings can be found in the article by Hertz, Shirk, Ricketts, and Weisshaar[7]. Still, there is only a very limited amount of wind tunnel data available which examines the effect of structural bending-torsion coupling on flutter speed or that verifies the accuracy of analytical predictions. Additionally, the phenomenon of stall flutter has received only limited attention as it relates to metal lifting surfaces[8,9], and virtually none as related to advance composite lifting surfaces.

To add to the limited experimental data on aeroelastic tailoring with graphite/epoxy, an investigation was conducted to determine the effect of bending-torsion stiffness coupling on both the divergence and flutter velocities of unswept lifting surfaces in incompressible flow. Types of flutter examined included both potential flow (low angle of attack) and stall flutter. This paper summarizes the analysis, design, and testing of six wind tunnel models with widely varying amounts of bending-torsion coupling. The models were of aircraft wings approximated by cantilevered, rectangular, flat plates constructed of graphite/epoxy with half span aspect ratios of four (eight, by standard aerodynamic convention). The present paper is based on an M.S. thesis by the first author[10].

## 2. Theory

### a) Anisotropic Plate Flexural Stiffness

The flexural modulus components $D_{ij}$ of a laminated advanced composite plate depend on both the fiber orientation and stacking sequence of the individual laminas. This development will consider only laminated plates with a mid-plane symmetric stacking sequence, due to the difficulties in fabricating flat unsymmetric laminated plates. The ply angles $\theta$ follow the sign convention in Fig. 1.

The in-plane, on-axis lamina modulus components $Q_{ij}$ were obtained from the orthotropic engineering constants for Hercules AS1/3501-6 unidirectional graphite/epoxy tape, from which the wind tunnel models were fabircated. These engineering constants take on different values depending on whether they are obtained from in-plane or out-of-plane dynamic tests. Turner[11] explored this subject in detail, and the engineering constants which

*Capt., USAF, Aeronautical Engineer, Member AIAA

**Professor of Aeronautics and Astronautics, Member AIAA he derived from out-of-plane, dynamic loads were used for the analysis results presented in this paper. For comparison purposes, Table 1 presents the orthotropic engineering constants derived by Turner, along with standard values used by Massachusetts Institute of Technology (MIT) which were obtained from in-plane tests.

Table 1 Orthotropic Engineering Constants

| Property: (Hercules AS1/3501-6 Graphite/Epoxy) | In-Plane Loading | Out-of-Plane Loading |
|---|---|---|
| $E_L$ | $130 \times 10^9$ N/m$^2$ | $98 \times 10^9$ N/m$^2$ |
| $E_T$ | $10.5 \times 10^9$ N/m$^2$ | $7.9 \times 10^9$ N/m$^2$ |
| $\nu_{LT}$ | 0.28 | 0.28 |
| $G_{LT}$ | $6.0 \times 10^9$ N/m$^2$ | $5.6 \times 10^9$ N/m$^2$ |

Ply thickness, $t_p$ $.134 \times 10^{-3}$ m

Density, $\rho_G$ 1520 kg/m$^3$

The flexural modulus $D_{ij}$ for an n-ply laminate with arbitrary ply angle orientation was obtained from the expression,[12]

$$D_{ij} = \sum_{k=1}^{n} Q_{ij}^{(\theta_k)} [z_k^3 - z_{k-1}^3]/3 \quad i,j = 1,2,6 \qquad (1)$$

where

$Q_{ij}^{(\theta_k)}$ = off-axis lamina modulus of $k^{th}$ ply

$\theta_k$ = ply angle of the $k^{th}$ ply

$z_k$ = distance up from the mid-plane to the upper surface of the $k^{th}$ ply

Table 2 presents the flexural moduli $D_{ij}$ as computed for the six laminates used in the present investigation.

Table 2 Flexural Moduli for Laminates

| Laminate | $D_{11}$ | $D_{12}$ | $D_{16}$ | $D_{22}$ | $D_{26}$ | $D_{66}$ |
|---|---|---|---|---|---|---|
| | | | (all values in N-m) | | | |
| $[0_2/90]_s$ | 4.125 | .960 | 0 | .490 | 0 | .243 |
| $[\pm45/0]_s$ | 1.550 | .928 | .437 | 1.404 | .437 | 1.075 |
| $[+45_2/0]_s$ | 1.550 | .928 | .946 | 1.404 | .946 | 1.075 |
| $[-45_2/0]_s$ | 1.550 | .928 | -.946 | 1.404 | -.946 | 1.075 |
| $[+30_2/0]_s$ | 2.704 | .720 | 1.180 | .666 | .459 | .866 |
| $[-30_2/0]_s$ | 2.704 | .720 | -1.180 | .666 | -.459 | .866 |

## b) Rayleigh-Ritz Formulation

The direct Rayleigh-Ritz energy method is a relatively simple, straight forward appproximation for plate deflections, as required for the free vibration, flutter, and divergence analyses in this paper. The "wing" was idealized by a rectangular cantilevered flat plate with uniform thickness.

The Rayleigh-Ritz analysis begins by assuming a deflection shape for the structure. Since the first three vibration modes would probably be of most interest for flutter analysis, a five term deflection equation was chosen. These five terms approximated the plate deflection for the first bending (1B), second bending (2B), first torsion (1T), second torsion (2T), and first chordwise (1C) vibration modes. Originally, only the first three terms were taken, but later it was shown by Jensen,[13] that the last two terms were necessary to improve the accuracy of the approximations for the first three vibration modes. The deflection equation, written in generalized coordinates is:

$$w = \sum_{i=1}^{5} \gamma_i \; (x,y) \; q_i(t) \qquad (2)$$

where w is the lateral deflection, $\gamma_i$ (x,y) is the nondimensional deflection or mode shape of the $i^{th}$ mode, and $q_i(t)$ is the generalized displacement of the $i^{th}$ mode. The mode shapes all satisy the geometric boundary conditions for a cantilevered plate, and the generalized displacement has units of length and is a function only of time. For convenience and consistency, the sign conventions in Fig. 1 were used for development of all analysis methods.

The mode shape can be written as:

$$\gamma_i \; (x,y) = \phi_i \; (x) \; \psi_i \; (y) \qquad i = 1 \ldots 5 \qquad (3)$$

where $\phi$ and $\psi$ are single dimension mode shapes in the x and y directions, respectively.

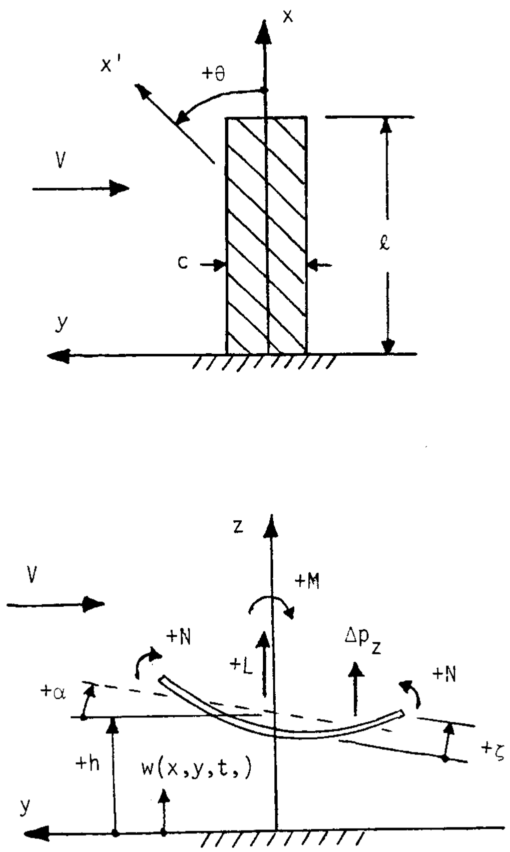


Fig. 1 Plate Layout and Sign Conventions

The mode shapes used in this paper are shown below.

$$
\begin{aligned}
\phi_1(x) &= 1^{st}\ \text{cantilever beam mode} & \psi_1(y) &= 1\\
\phi_2(x) &= 2^{nd}\ \text{cantilever beam mode} & \psi_2(y) &= 1\\
\phi_3(x) &= \sin\left(\frac{\pi x}{2\ell}\right) & \psi_3(y) &= \frac{y}{c}\\
\phi_4(x) &= \sin\left(\frac{3\pi x}{2\ell}\right) & \psi_4(y) &= \frac{y}{c}\\
\phi_5(x) &= \frac{x}{\ell}\left(1-\frac{x}{\ell}\right) & \psi_5(y) &= \left[4\left(\frac{y}{c}\right)^2-\frac{1}{3}\right]
\end{aligned}
\tag{3a}
$$

The strain energy U for a symmetric anisotropic laminate is[14]

$$
\begin{aligned}
U = \frac{1}{2}\int_0^{\ell}\int_{-\frac{c}{2}}^{\frac{c}{2}} &[D_{11}(w,_{xx})^2 + 2D_{12}w,_{xx}w,_{yy}\\
&+ D_{22}(w,_{yy})^2 + 4D_{16}w,_{xx}w,_{xy} + 4D_{26}w,_{yy}w,_{xy}\\
&+ 4D_{66}(w,_{xy})^2]\,dydx
\end{aligned}
\tag{4}
$$

where a comma denotes partial differentiation.

Taking the partial derivatives of the deflection equation and using summation notation, Eq. (4) becomes

$$U = \frac{1}{2}\sum_{i=1}^{5}\sum_{j=1}^{5} K_{ij}\, q_i\, q_j \tag{5}$$

where $K_{ij}$ is an element of a symmetric 5 x 5 matrix, defined as

$$
\begin{aligned}
K_{ij} = \int_0^{\ell}\int_{-\frac{c}{2}}^{\frac{c}{2}} \Big\{ &D_{11}(\gamma_i),_{xx}(\gamma_j),_{xx} + D_{22}(\gamma_i),_{yy}(\gamma_j),_{yy}\\
&+4D_{66}(\gamma_i),_{xy}(\gamma_j),_{xy} + D_{12}[(\gamma_i),_{xx}(\gamma_j),_{yy}\\
&\quad + (\gamma_i),_{yy}(\gamma_j),_{xx}]\\
&+ 2D_{16}[(\gamma_i),_{xx}(\gamma_j),_{xy} + (\gamma_i),_{xy}(\gamma_j),_{xx}]\\
&+ 2D_{26}[(\gamma_i),_{yy}(\gamma_j),_{xy} + (\gamma_i),_{xy}(\gamma_j),_{yy}]\Big\}\,dydx
\end{aligned}
\tag{5a}
$$

The kinetic energy expression for the plate is

$$T = \frac{1}{2}\int_0^{\ell}\int_{-\frac{c}{2}}^{\frac{c}{2}} m\,\dot{w}^2\,dydx \tag{6}$$

where $(\dot{\ })$ = d/dt and m is the mass/area, $m = \rho_G\, t_G$, where $\rho_G$ is the density of graphite/epoxy and $t_G$ is the total plate thickness. Rewriting Eq. (6) using summation notation yields

$$T = \frac{1}{2}\sum_{i=1}^{5}\sum_{j=1}^{5} M_{ij}\,\dot{q}_i\,\dot{q}_j \tag{7}$$

where $M_{ij}$ is an element of 5 x 5 matrix defined as

$$M_{ij} = \int_0^{\ell}\int_{-\frac{c}{2}}^{\frac{c}{2}} \gamma_i\, m\,\gamma_j\, dy\, dx \tag{7a}$$

The change in external work $\delta W_e$ can be expressed as

$$\delta W_e = \int_0^{\ell}\int_{-\frac{c}{2}}^{\frac{c}{2}} \Delta p_z\, \delta w\, dydx \tag{8}$$

where $\Delta p_z$ is a distributed lateral load per unit area. Again, expressing Eq. (8) using summation notation yields

$$\delta W_e = \sum_{i=1}^{5} Q_i\, \delta q_i \tag{9}$$

where $Q_i$ is a generalized force defined as

$$Q_i = \int_0^{\ell}\int_{-\frac{c}{2}}^{\frac{c}{2}} \Delta p_z\, \gamma_i\, dydx \tag{9a}$$

The relationship between the work and energy expressions is obtained through Lagrange's equation, which is a statement of Hamilton's energy principle; the basic premise of the Rayleigh-Ritz method. Lagrange's equation is[15]

$$\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_i}\right) - \frac{\partial T}{\partial q_i} + \frac{\partial U}{\partial q_i} = Q_i \qquad (i = 1,2,\ldots 5) \tag{10}$$

Taking the partial derivatives results in five equations of motion, expressed here in matrix form

$$[M]\ddot{\underset{\sim}{q}} + [K]\underset{\sim}{q} = \underset{\sim}{Q} \qquad (i,j = 1\ldots 5) \tag{11}$$

where the elements of the diagonal mass matrix [M] are

$$M_{11} = mc\int_0^{\ell} \phi_1^2\,dx \qquad M_{22} = mc\int_0^{\ell} \phi_2^2\,dx$$

$$M_{33} = \frac{mc}{12}\int_0^{\ell} \phi_3^2\,dx \qquad M_{44} = \frac{mc}{12}\int_0^{\ell} \phi_4^2\,dx$$

$$M_{55} = \frac{4mc}{45}\int_0^{\ell} \phi_5^2\,dx \tag{11a}$$

and the elements of the symmetric stiffness matrix [K] are:

$$K_{11} = D_{11}c\int_0^{\ell} \phi_1''^2\,dx \qquad K_{12} = 0$$

$$K_{13} = 2D_{16}\int_0^{\ell} \phi_1''\,\phi_3'\,dx \qquad K_{14} = 2D_{16}\int_0^{\ell} \phi_1''\,\phi_4'\,dx$$

$$K_{15} = \frac{8D_{12}}{c}\int_0^{\ell} \phi_1''\,\phi_5\,dx \qquad K_{22} = D_{11}c\int_0^{\ell} \phi_2''^2\,dx$$

$$K_{23} = 2D_{16}\int_0^{\ell} \phi_2''\,\phi_3'\,dx \qquad K_{24} = 2D_{16}\int_0^{\ell} \phi_2''\,\phi_4'\,dx$$

$$K_{25} = \frac{8D_{12}}{c}\int_0^{\ell} \phi_2''\,\phi_5\,dx$$

$$K_{33} = \left[\frac{4D_{66}}{c}\int_0^{\ell} (\phi_3')^2\,dx\right]\left(\frac{k_{1T}}{1.57}\right)^2$$

$$K_{34} = 0$$

$$K_{35} = \frac{4}{3}D_{16}\int_0^{\ell} \phi_3''\,\phi_5'\,dx + \frac{16D_{26}}{c^2}\int_0^{\ell} \phi_3'\,\phi_5\,dx$$

$$K_{44} = \left[\frac{4D_{66}}{c}\int_0^{\ell} \phi_4'^2\,dx\right]\left(\frac{k_{2T}}{4.71}\right)^2$$

$$K_{45} = \frac{4}{3}D_{16}\int_0^{\ell} \phi_4''\,\phi_5'\,dx + \frac{16D_{26}}{c^2}\int_0^{\ell} \phi_4'\,\phi_5\,dx$$

$$K_{55} = \frac{4}{45}D_{11}c\int_0^{\ell} (\phi_5'')^2\,dx + \frac{64}{3}\frac{D_{66}}{c}\int_0^{\ell} (\phi_5')^2\,dx + \frac{64D_{22}}{c^3}\int_0^{\ell} (\phi_5)^2\,dx \tag{11a}$$

The integrals were non-dimensionalized and solved numerically using 100 point trapezoidal integration. The terms $k_{1T}$ and $k_{2T}$ are structural aspect ratio corrections developed by Crawley and Dugundji[16] and presented for the laminates of interest, with $\ell/c = 4$, in Table 3.

Table 3 Structural Aspect Ratio Corrections

| Laminate | $\beta$ | $k_{1T}$ | $k_{2T}$ |
|---|---|---|---|
| $[0_2/90]_s$ | .0221 | 1.87 | 6.30 |
| $[\pm 45/0]_s$ | .0019 | 1.63 | 5.00 |
| $[+45_2/0]_s$, $[-45_2/0]_s$ | .0019 | 1.63 | 5.00 |
| $[+30_2/0]_s$, $[-30_2/0]_s$ | .0041 | 1.68 | 5.20 |

Note: $\beta = D_{11}/48\,D_{66}\,(\frac{\ell}{c})^2$

c) Free Vibration Analysis

The free vibration problem was formulated by setting the generalized forces $Q_i$ equal to zero in Eq. (11). Assuming harmonic (sinusoidal) motion, the generalized displacements can be expressed as

$$\underset{\sim}{q} = \underset{\sim}{q}e^{i\omega t} \qquad \ddot{\underset{\sim}{q}} = -\omega^2\,\underset{\sim}{q}e^{i\omega t} \tag{12}$$

where $\omega$ is the frequency. These expressions, when substituted into the differential equations of motion, Eq. (11), results in the real eigenvalue problem,

$$\{[K] - \omega^2\,[M]\}\,\underset{\sim}{q} = 0 \tag{13}$$

Eq. (13) is solved for the eigenvalues $\omega_i^2$ and associated eignevectors $\underset{\sim}{q}_i$. Then the natural vibration frequencies $\omega_i$ are found from $\omega_i = \sqrt{\omega_i^2}$, while the associated vibration mode shapes are found by substituting $\underset{\sim}{q}_i$ back into Eq. (2).

d) Aeroelastic Flutter Analysis

The flutter analysis was done using the familiar V-g method.[15] In this method, the structural damping coefficient g, introduced into the equations of motion, is plotted versus velocity for each vibration mode. Since solutions to the equations of motion represent conditions for neutral stability, the value of g obtained in this manner represents the amount of damping that must be added to the structure to attain neutral stability (flutter) at the given velocity. Thus, negative values of structural damping indicate that the structure is stable. Flutter occurs when the structural damping equals the actual damping of the structure.

Assuming harmonic motion, the equations of motion are,

$$\{[K] - \omega^2 [M]\} \underset{\sim}{q} = \underset{\sim}{Q}/e^{i\omega t} \tag{14}$$

where $\underset{\sim}{Q}$ represents the aerodynamic forces on the plate and is given generally by Eq. (9a). For the lateral deflection modes given by Eqs. (2), (3), and (3a), these aerodynamic forces become

$$\begin{aligned} Q_i &= \int_0^\ell \phi_i \, L \, dx \qquad (i = 1,2) \\ &= \frac{1}{c} \int_0^\ell \phi_i \, M \, dx \qquad (i = 3,4) \\ &= \int_0^\ell \phi_5 \, N \, dx \qquad (i = 5) \end{aligned} \tag{15}$$

where L, M, and N represent the lift, moment, and camber force about the midchord, namely,

$$L = \int_{-\frac{c}{2}}^{\frac{c}{2}} \Delta p_z \, dy, \quad M = \int_{-\frac{c}{2}}^{\frac{c}{2}} y \, \Delta p_z \, dy,$$

$$N = \int_{-\frac{c}{2}}^{\frac{c}{2}} \psi_5 \, \Delta p_z \, dy \tag{16}$$

Using 2-dimensional, incompressible, unsteady aerodynamic theory and including the camber effects as adapted from Spielberg[17], the aerodynamic forces can be expressed as,

$$\begin{aligned} L &= \pi\rho\omega^2 b^3 e^{i\omega t} \left(L_A \frac{h}{b} + L_B \alpha + L_C \frac{\zeta}{b}\right) \\ M &= \pi\rho\omega^2 b^4 e^{i\omega t} \left(M_A \frac{h}{b} + M_B \alpha + M_C \frac{\zeta}{b}\right) \\ N &= \pi\rho\omega^2 b^3 e^{i\omega t} \left(N_A \frac{h}{b} + N_B \alpha + N_C \frac{\zeta}{b}\right) \end{aligned} \tag{17}$$

where the deflections and angles at the midchord h, $\alpha$, and $\zeta$, above are related to the deflection shapes Eqs. (2), (3), (3a) through

$$\begin{aligned} h &= \phi_1 (x) \, q_1 + \phi_2(x) \, q_2 \\ \alpha &= \frac{1}{c} [\phi_3 (x) \, q_3 + \phi_4 (x) \, q_4] \\ \zeta &= \phi_5 (x) \, q_5 \end{aligned} \tag{18}$$

See Fig. 1 for positive directions of all deflections and forces. In Eq. (17) $\rho$ is the air density, b is the semichord, $\omega$ is the oscillation frequency, and the functions $L_A$, $L_B$, $L_C$, $M_A$,... are non-dimensional complex functions of reduced frequency $k = \omega b/V$, given by,

$$\begin{aligned} L_A &= 1 - i \frac{2C(k)}{k} \\ L_B &= \frac{i}{k} + i \frac{C(k)}{k} + \frac{2C(k)}{k^2} \\ L_C &= -\frac{1}{12} - i \frac{C(k)}{3k} - \frac{2C(k)}{k^2} \end{aligned} \tag{19}$$

$$\begin{aligned} M_A &= -i \frac{C(k)}{k} \\ M_B &= \frac{1}{8} - \frac{i}{2k} + i \frac{C(k)}{2k} + \frac{C(k)}{k^2} \\ M_C &= \frac{i}{2k} - i \frac{C(k)}{6k} + \frac{1}{k^2} - \frac{C(k)}{k^2} \end{aligned} \tag{19}$$

$$\begin{aligned} N_A &= -\frac{1}{12} - i \frac{C(k)}{3k} \\ N_B &= -\frac{i}{3k} + i \frac{C(k)}{6k} + \frac{C(k)}{3k^2} \\ N_C &= \frac{1}{36} - i \frac{C(k)}{18k} + \frac{1}{2k^2} - \frac{C(k)}{3k^2} \end{aligned}$$

where C(k) is the Theodorsen function. Finally, combining Eqs. (15), (17), and (18), the generalized forces $Q_i$ can be written conveniently in summation notation as,

$$Q_i = \pi\rho\omega^2 b^3 e^{i\omega t} \sum_{j=1}^{5} A_{ij} \, q_j \tag{20}$$

where $A_{ij}$ are the elements of the 5 x 5 aerodynamic matrix [A],

$$\begin{aligned}
A_{11} &= L_A \frac{1}{b} \int_0^\ell \phi_1^2 \, dx & A_{12} &= 0 \\
A_{13} &= L_B \frac{1}{c} \int_0^\ell \phi_1\phi_3 \, dx & A_{14} &= L_B \frac{1}{c} \int_0^\ell \phi_1\phi_4 \, dx \\
A_{15} &= L_C \frac{1}{b} \int_0^\ell \phi_1\phi_5 \, dx & A_{21} &= 0 \\
A_{22} &= L_A \frac{1}{b} \int_0^\ell \phi_2^2 \, dx & A_{23} &= L_B \frac{1}{c} \int_0^\ell \phi_2\phi_3 \, dx \\
A_{24} &= L_B \frac{1}{c} \int_0^\ell \phi_2\phi_4 \, dx & A_{25} &= L_C \frac{1}{b} \int_0^\ell \phi_2\phi_5 \, dx \\
A_{31} &= M_A \frac{1}{c} \int_0^\ell \phi_3\phi_1 \, dx & A_{32} &= M_A \frac{1}{c} \int_0^\ell \phi_3\phi_2 \, dx \\
A_{33} &= M_B \frac{b}{c^2} \int_0^\ell \phi_3^2 \, dx & A_{34} &= 0 \\
A_{35} &= M_C \frac{1}{c} \int_0^\ell \phi_3\phi_5 \, dx & A_{41} &= M_A \frac{1}{c} \int_0^\ell \phi_4\phi_1 \, dx \\
M_{42} &= M_A \frac{1}{c} \int_0^\ell \phi_4\phi_2 \, dx & A_{43} &= 0 \\
A_{44} &= M_B \frac{b}{c^2} \int_0^\ell \phi_4^2 \, dx & A_{45} &= M_C \frac{1}{c} \int_0^\ell \phi_4\phi_5 \, dx \\
A_{51} &= N_A \frac{1}{b} \int_0^\ell \phi_5\phi_1 \, dx & A_{52} &= N_A \frac{1}{b} \int_0^\ell \phi_5\phi_2 \, dx \\
A_{53} &= N_B \frac{1}{c} \int_0^\ell \phi_5\phi_3 \, dx & A_{54} &= N_B \frac{1}{c} \int_0^\ell \phi_5\phi_4 \, dx \\
& & A_{55} &= N_C \frac{1}{b} \int_0^\ell \phi_5^2 \, dx
\end{aligned} \tag{20a}$$

The integrals were again nondimsionalized and evaluated numerically.

Using Eqs. (14) and (20), the equations of motion for the flutter analysis are written in matrix notation as,

$$[K]\,\underset{\sim}{q} - \omega^2\,[M]\,\underset{\sim}{q} = \pi\rho\omega^2\,b^3\,[A]\,\underset{\sim}{q} \tag{21}$$

where,

$$[B] = [M] + \pi\rho b^3\,[A] \tag{22}$$

Structural damping is introduced by multiplying [K] by (i + ig).[15] A complex eigenvalue Z defined as,

$$Z = \frac{1 + ig}{\omega^2} \tag{23}$$

allows the equations of motion to be written as a standard complex eigenvalue problem,

$$\{[B] - Z\,[K]\}\,\underset{\sim}{q} = 0 \tag{24}$$

Choosing a value of reduced frequency k, the aerodynamic coefficients [A] are evaluated, and Eq. (24) is solved for the five complex eigenvalues $Z_i$. For each eigenvalue $Z_i$, the frequency $\omega$, the structural damping g, and the velocity V required for flutter are formed through the relations,

$$\omega = \frac{1}{\sqrt{Re(Z)}}\,,\quad g = \frac{Im(Z)}{Re(Z)}\,,\quad V = \frac{b\omega}{k} \tag{25}$$

The procedure is repeated with new k values, to obtain the V-g diagram, created by plotting the structural damping for each eigenvalue versus velocity. The flutter velocity $V_F$ was conservatively chosen to be the point where the damping coefficient g from any root first crosses the V-axis, and the associated value of $\omega$ is the flutter frequency $\omega_F$. To simplify calculation of the Theodorsen function C(k), the R. T. Jones approximation[15] was used, namely,

$$C(p) = \frac{0.5\,p^2 + 0.2808\,p + 0.01365}{p^2 + 0.3455\,p + 0.01365} \tag{26}$$

where p = ik.

The flutter analysis technique involving the creation of a V-g diagram is computationally intensive and best implemented on a digital computer. Since [B] is complex and [K] is symmetric, a standard FORTRAN subroutine was chosen from the International Mathematical and Statistical Libraries (IMSL) which first converted [B] to upper Hessenberg form and [K] to upper triangular form, then extracted the eigenvalues. The output of the computer program was a tabular listing (for many values of k) of $\omega$, g, and V corresponding to each of the five eigenvalues. The V-g diagrams were then plotted manually. The natural frequencies were also obtained from this computer program by inserting a small value for the air density and a large value for k. This corresponded to the vibration frequency at $V \approx 0$ in a vacuum. Normal values of air density with large values of k give frequencies with air mass effects present.

e) Divergence Analysis

Divergence is a static deflection phenomenon in which the aerodynamic forces on a wing exceed the resisting or restoring forces of the structure, often resulting in catostrophic failure of the wing. The divergence velocity can also be determined from the V-g diagram, and is characterized by the structural damping of a deflection branch abruptly going from a negative value to zero (but not crossing the V-axis) at the same time as the frequency goes to zero.

For divergence analysis, the infinite span lift curve slope $\frac{\partial C\ell}{\partial\alpha} = 2\pi$ was empirically corrected to account for the effects of finite span by applying the overall correction suggested in Reference 15,

$$\frac{\partial C\ell}{\partial\alpha} = 2\pi\,\left[\frac{AR}{AR+2}\right] \tag{27}$$

where AR is the aspect ratio $2\ell/c$ of the wing. This is equivalent to modifying the divergence velocity $V_D$ found from the $2\pi$ lift curve slope as shown below,

$$(V_D)_{AR} = \sqrt{\frac{AR+2}{AR}}\,(V_D)_{2\pi} \tag{28}$$

The above aspect ratio correction was not used for the flutter speed calculations.

Alternatively for the divergence calculations, the divergence speed could be calculated directly by neglecting the mass terms [M] in Eq. (21), and writing the resulting equations in the form,

$$[K]\,\underset{\sim}{q} = \pi\rho V^2\,b\,k^2\,[A]\,\underset{\sim}{q} \tag{29}$$

This results in the standard real eigenvalue problem,

$$\{[A^{stat}] - \lambda\,[K]\}\,\underset{\sim}{q} = 0 \tag{30}$$

where,

$$[A^{stat}] = \lim_{k \to 0}\left\{k^2\,[A]\right\} \tag{30a}$$

$$\lambda = 1/\pi\rho\,V^2 b$$

The largest positive $\lambda$ gives the lowest divergence speed, $V_D$. Again, this $V_D$ should be corrected for finite aspect ratio by applying Eq. (28).

## 3. Experiment

### a) Model Design and Fabrication

The wind tunnel model was conceived as a half-span representation of an aircraft wing using a plate cantilevered at the base. Model design was driven by several constraints imposed by the study objectives and available facilities. The models had to,

(1) demonstrate a wide range of bending-torsion coupling;

(2) be rectangular shaped constant thickness, flat plates, with zero sweep;

(3) exhibit flutter within the 32 m/sec velocity limitation of the wind tunnel;

(4) be small enough to be constructed with the available equipment for cutting and curing graphite/epoxy at MIT;

(5) be tough enough to withstand repeated large static and oscillatory loads.

Constraints (3) and (4) directly opposed one another and drove the design to a very thin laminate, which increased the possibility of warpage during curing. Constraint (1) seemed to indicate that an unbalanced but mid-plane symmetric lamination was desirable. This also increased the possibility of warpage during curing.

The total thickness finally chosen was six plies. Five different laminates were selected as representative of a wide range of bending-torsion coupling stiffnesses, both positive and negative. The $[+30_2/0]_s$ and $[-30_2/0]_s$ laminates were chosen because they had the highest theoretical bending-torsion stiffness $D_{16}$. The $[+45_2/0]_s$ and $[-45_2/0]_s$ laminates were chosen because they had the highest theoretical torsion stiffness $D_{66}$. The $[0_2/90]_s$ laminate was chosen for its zero $D_{16}$. During the test program, another laminate, $[+45/0]_s$, was added because it had the same theoretical $D_{11}$ and $D_{66}$ as $[+45_2/0]_s$ with less $D_{16}$. To minimize the number of test specimens which had to be constructed, the $[+\theta_2/0]_s$ and $[-\theta_2/0]_s$ test specimens were actually the same laminate, simply rotated 180 degrees about the x-axis (see Fig. 1).

Each model had an overall length of 330 mm (13 in) and a chord of 76 mm (3 in). The overall length included a 25 mm (1 in) loading tab, making the effective cantilever plate length 305 mm (12 in). This gave the plate an aspect ratio ($\ell/c$) of four. Finally, in an effort to minimize plate stiffness, an airfoil shaped fairing was not used for these wind tunnel tests.

Individual plys were manually cut from Hercules AS1/3501-6 graphite/epoxy prepreg tape and layed up into 305 mm by 356 mm laminates, which were cured in an autoclave. Models were cut from the laminats using a diamond cutting wheel mounted on an automatic feed milling machine. Loading tabs were machined from 2.4 mm aluminum plate and epoxied to the base of each model. A bending and a torsion strain gauge were attached to the base of both sides of each model, at the midchord. The guages were wired to compensate for temperature changes during tests. Prior to attaching the strain guages and loading tabs, each model was measured (thickness, width, and length), weighed, and checked for warping. These figures were recorded and compared to nominal (theoretical) values.

### b) Vibration Test Apparatus

A vise machined from a 25.4 mm x 152 mm x 29 mm aluminum block was suspended by four spring steel strips, allowing it to translate in the Z-direction (Fig. 1); but restricting motion in all other directions. This vise was attached to an electromagnetic shaker driven by an audio amplifier. A frequency generator with a digital readout provided the imput signal. The shaker had a rated frequency range of 5 to 3000 Hz.

### c) Wind Tunnel Test Apparatus

All wind tunnel tests were performed in the MIT Department of Aeronatuics and Astronautics, acoustic wind tunnel. The acoustic wind tunnel was a continuous flow tunnel with a 1.5 m (5 ft) by 2.3 m (7.5 ft) free jet test section 2.3 m (7.5 ft) long, located inside a large anechoic chamber. The tunnel had a velocity range of 0 to 32 m/sec (0 to 105 ft/sec), which was read from an alcohol manometer.

The test apparatus (see Fig. 2) consisted of a turntable machined from aluminum and mounted on a 6.35 mm tall pedestal made of 152 mm steel pipe. The pedestal was mounted directly to a rigid platform in the floor of the test section. The vise from the free vibration test apparatus was fastened to the turntable and was covered by a wooden disk to provide smooth airflow past the model. Angle of attack graduations (2° increments) were marked on the disk and the turntable was rotated during the tests by an aluminum rod extending outside the test section. The model was mounted in a vertical position to eliminate any gravity effects. The instrumentation system was a Gould 2400 series four channel stripchart recorder, which had an internal time base generator, and recorded the bending and torsion strains of the model.

### d) Test Procedures

The first three vibration modes were obtained by slowly increasing the frequency and visually identifying when the amplitude of each mode peaked. The frequency at this condition was then read from the counter. The free vibration tests were repeated several times to insure consistent results.

The wind tunnel tests began by exciting the first bending and torsion modes of the model under zero velocity conditions. The resulting damped oscillations were recorded on the stripchart. The model was set at zero angle of attack and the wind tunnel run up and stabilized at the first velocity. The angle of attack was then varied from +18° to -18°, with a stripchart record being made at 2° increments. The model was reset to zero angle of attack, the velocity increased, and after it stabilized, the angle of attack sweep was repeated. This procedure was repeated until the model either fluttered or diverged at zero angle of attack, or the maximum speed of the wind tunnel was reached.

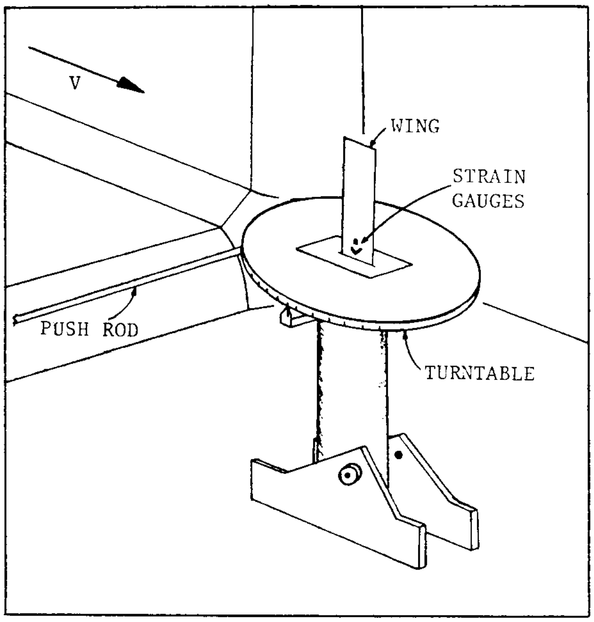


Fig. 2   Test Set-up in Wind Tunnel

Divergence testing often results in the destruction of a model. Since this was not a desirable result, wind tunnel speeds were increased in very small increments as the divergence velocity was approached. The model deflections were closely monitored, and when it would no longer return to a zero deflection state, the tunnel velocity was not increased further. Because the models were rather flexible, they proved very robust and after many tests, not a single one was lost.

All data reduction was done manually (with the aid of a programmable calculator). Flutter frequencies were annotated on the strip chart, along with the velocity. On selected strip chart records, peak-to-peak amplitudes of both bending and torsion strain gauges were annotated. Finally, the flutter boundary velocities were plotted versus angle of attack for each model.

## 4. Discussion of Results

### a) Free Vibration Tests

Table 4 gives the experimental and calculated natural frequencies of the six laminates tested. Comparing the experimental frequencies of the balanced laminate $[\pm45/0]_s$ and the unbalanced laminate $[+45_2/0]_s$, one observes that the frequencies of the first three vibration modes were lower for the unbalanced laminate. Since the theoretical bending stiffness $D_{11}$, torsion stiffness $D_{66}$, and chordwise stiffness $D_{22}$, were the same for both of these models, (see Table 2), the differences in frequencies could be attributed entirely to the coupling stiffness $D_{16}$ and $D_{26}$. This stiffness coupling apparently reduces both the bending and torsion frequencies of the wing.

As shown in Table 4, the Rayleigh-Ritz analysis using the five modes given in Eqs (2), (3) and (3a) gave generally good agreement with experiment.

Table 4   Natural Frequencies of Models

| Model | Vibration Mode | Exper. Freq. (Hz) | Calc. Freq. (Hz) (5 modes) | Calc. Freq. (Hz) (3 modes) |
|---|---|---|---|---|
| $[0_2/90]_s$ | 1B | 11.1 | 10.7 | 11.1 |
| | 1T | 42 | 39 | 33 |
| | 2B | 69 | 67 | 69 |
| | 2T | | 132 | |
| | 1C | | 430 | |
| $[\pm45/0]_s$ | 1B | 6.1 | 5.7 | 6.4 |
| | 2B | 38 | 37 | 42 |
| | 1T | 77 | 69 | 70 |
| | 2T | | 216 | |
| | 1C | | 746 | |
| $[+45_2/0]_s$ $[-45_2/0]_s$ | 1B | 4.8 | 4.6 | 4.7 |
| | 2B | 30 | 32 | 41 |
| | 1T | 51 | 55 | 71 |
| | 2T | | 205 | |
| | 1C | | 751 | |
| $[+30_2/0]_s$ $[-30_2/0]_s$ | 1B | 6.0 | 6.0 | 5.8 |
| | 2B | 36 | 41 | 52 |
| | 1T | 58 | 60 | 66 |
| | 2T | | 196 | |
| | 1C | | 542 | |

These were obtained by using the V-g method at large values of k, and hence had a small correction for virtual air mass present also. As mentioned earlier, a previous analysis which included only the first three modes 1B, 2B, and 1T, did not predict the coupled 2B and coupled 1T modes adequately. For interest, these results are also shown in Table 4. This vibration problem was later resolved by Jensen,[13] and led to the present five mode formulation.

### b) Wind Tunnel Tests

The primary objective of the wind tunnel tests was to obtain zero angle of attack flutter and divergence velocities for each model, which could be compared to analytical results. The secondary objective was to examine the stall flutter behavior of each model. Since the test velocities were low and the models quite flexible, the onset of flutter at zero angle of attack tended to be gradual. For consistency, the flutter threshold was defined to be the velocity where the oscillation amplitude stabilized and the waveform became periodic. Determining the threshold for stall flutter was somewhat easier, becuase oscillation tended to begin abruptly when the stall angle of attack was reached.

The type of flutter (bending, torsion, or bending-torsion) could not be determined completely from the amplitude of the bending and torsion strip chart channels, since for a plate with bending-torsion coupling, the bending and torsion motions are mixed together. Accordingly, it was decided to define bending flutter as oscillations with a frequency very close to one of the bending natural frequencies, torsion flutter when the frequency was close to the torsion natural frequency, and bending-torsion flutter when the frequency was between the two.

The flutter threshold velocity is plotted versus initial root angle of attack in Fig. 3. The $[0_2/90]_s$ model, with no bending-torsion coupling, exhibited bending-torsion flutter at low angles of attack and torsion stall flutter at higher angles of attack. This behavior is similar to that reported by Rainey.[8] The models with large positive bending-torsion coupling, $[+30_2/0]_s$ and $[+45_2/0]_s$, exhibited primarily bending-torsion flutter. The flutter velocity here did not drop significantly with increasing intial angle of attack, probably because the coupling caused a decrease in the tip angle of attack, preventing it from stalling. At an initial angle of attack of 18 degrees, however, the flutter speed for the $[+30_2/0]_s$ model did drop significantly and the flutter changed to torsion stall flutter. The $[\pm 45_2/0]_s$ model, which was stiff in torsion and had only a small amount of positive bending-torsion coupling, would not flutter at zero angle of attack within the 32 m/sec maximum speed for the wind tunnel. However, at angles of attack above eight degrees, it behaved similarly to the $[0_2/90]_s$ model and exhibited torsion stall flutter. The models with large negative bending-torsion coupling, $[-30_2/0]_s$ and $[-45_2/0]_s$, exhibited divergence at zero angle of attack, and primarily bending stall flutter at initial angles of attack greater than one degree. This flutter was due to bending motion causing the blade to stall due to the bending-torsion stiffness coupling.

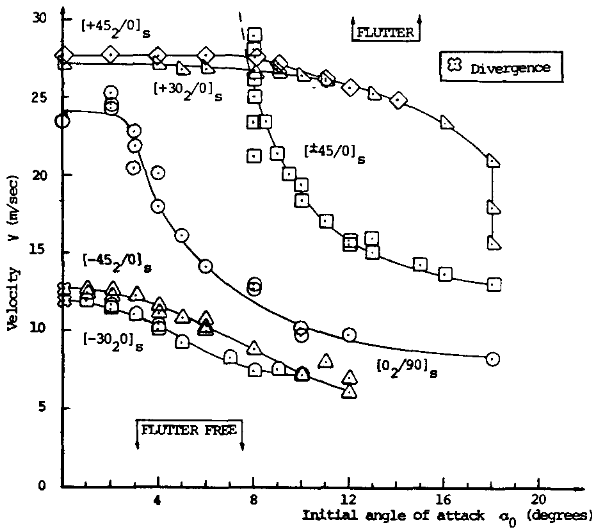


Fig. 3 Experimental Flutter and Divergence for All Plates

To illustrate more clearly how the direction of the angle plies determines the behavior of the model, wind tunnel data for the three models with 45 degree plies are presented in Fig. 4. These three models have vastly different aeroelastic behavior. The $[\pm 45/0]_s$ indicated torsion stall flutter, the $[+45_2/0]_s$ indicated bending-torsion flutter, and the $[-45_2/0]_s$ indicated divergence at zero angle of attack and bending stall flutter at higher angles of attack.

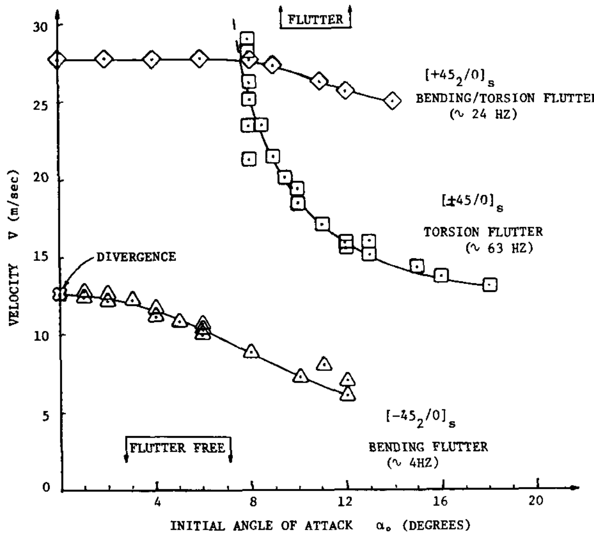


Fig. 4 Experimental Flutter and Divergence for 45° Ply Plates

The vastly different low angle of attack behavior observed in the wind tunnel data for these models was also evident in the theoretcal analysis. The V-g and V-ω diagrams are shown in Fig. 5 for the $[+45_2/0]_s$ model, the $[\pm 45/0]_s$ model, and the $[-45_2/0]_s$ modes. The torsion branch became unstable on the $[0_2/90]_s$ model and apparently on the $[\pm 45/0]_s$ model, while the second bending branch became unstable on the $[+45_2/0]_s$ model. The V-g and V-ω diagrams for the $[-45_2/0]_s$ model exhibited the classical first bending mode divergence. The remaining fourth and fifth mode branches are not shown in Fig. 5 since they always show a small stable negative g over the velocity range indicated. One should note that the V-g and V-ω diagrams are valid only for low angle of attack flutter or divergence.

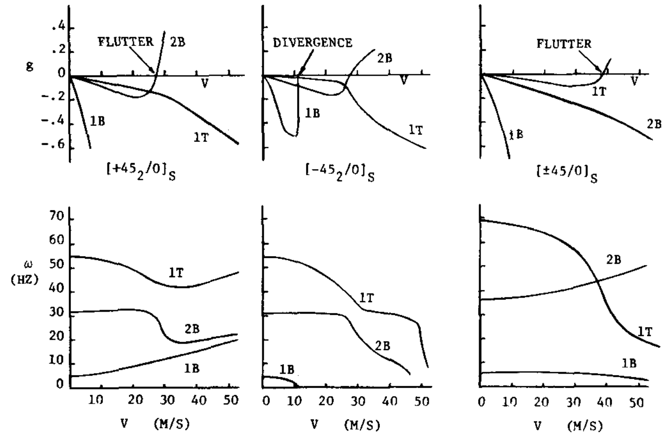


Fig. 5 Theoretical V-g Diagrams for 45° Ply Plates

The calculated and experimental zero angle of attack flutter velocities appear in Table 5. There was good agreement between analytical and experimental results for all cases in which a comparison could be made. The excellent agreement between analytical and experimental results for the $[+45_2/0]_s$ and $[+30_2/0]_s$ models is perhaps a bit fortuitous, but one may conclude that the analytical method appeared to provide a reasonable estimate for flutter velocities of the models in this paper. Both the experiments and analysis showed the flutter velocities of the $[45_2/0]_s$ and $[+30_2/0]_s$ models to be almost identical.

Table 5 Zero Angle of Attack Flutter Velocities

| Model | Flutter Velocity (m/sec) | | Flutter Frequency (Hz) | |
|---|---|---|---|---|
| | Experimental | Calc. | Experimental | Calc. |
| $[0_2/90]_s$ | 24 | 21.0 | ∿ 29 | 25 |
| $[\pm45/0]_s$ | > 32 | 39.0 | not obtained | 39 |
| $[+45_2/0]_s$ | 28 | 27.8 | ∿ 24 | 28 |
| $[-45_2/0]_s$ | divergence | 27.8 | divergence | 27 |
| $[+30_2/0]_s$ | 27 | 27.8 | ∿ 28 | 31 |
| $[-30_2/0]_s$ | divergence | 30.0 | divergence | 29 |

The calculated and experimental divergence velocities appear in Table 6. As with the flutter velocities, there was generally good agreement between experimental and analytical results (with finite span correction). Analysis results indicated that all models in this paper with positive stiffness coupling have an infinite divergence velocity. In examining the V-g diagram for the $(0_2/90]_s$ model, the authors observed that flutter occured at a velocity of 21 m/sec, followed closely by divergence at 25 m/sec. In the wind tunnel, the flutter occuring first probably obscured the divergence behavior. The $[-45_2/0]_s$ model had a slightly higher divergence velocity than the $[-30_2/0]_s$ model, while the analysis yielded a slightly opposite trend.

## 5. Conclusions

Bending-torsion coupling can be beneficial in delaying or eliminating divergence. This stiffness coupling can also be employed to delay the onset of stall flutter by causing tip stall of a wing to occur at a higher root angle of attack. The approximate analysis methods adapted for use in this paper predicted natural vibration frequencies, flutter velocities, and divergence velocities with reasonable accuracy, except for some slight differences in divergence speed trends for small variations in bending-torsion coupling. The analytical V-g

Table 6 Divergence Velocities

| Model | Divergence Velocity (m/sec) | | |
|---|---|---|---|
| | Experimental | Calc. ($C_{\ell_\alpha} = 2\pi$) | Calculated ($C_{\ell_\alpha} = 2\pi$ AR/AR+2) |
| $[0_2/90]_s$ | flutter | 22.3 | 25.0 |
| $[\pm45/0]_s$ | > 32.0 | infinite, no divergence | |
| $[+45_2/0]_s$ | flutter | infinite, no divergence | |
| $[-45_2/0]_s$ | 12.5 | 9.9 | 11.1 |
| $[+30_2/0]_s$ | flutter | infinite, no divergence | |
| $[-30_2/0]_s$ | 11.7 | 10.2 | 11.5 |

diagrams for these models showed that the presence of positive bending-torsion coupling switched the unstable flutter branch from torsion to second bending mode, while negative bending-torsion coupling caused a divergence instability in the first bending mode.

Wind tunnel tests on a series of rectangular straight wings varying bending-torsion coupling showed the widely varying aeroelastic behavior possible due to this stiffness coupling. Divergence instabilties and bending-torsion flutter occured for low angles of attack while torsion stall flutter and bending stall flutter occured at high angles of attack. There was reasonable agreement with theory at low angles of attack for the divergence and bending-torsion flutter. Torsion stall flutter occured on wings with little or positive bending-torsion coupling, while bending stall flutter occured on wings with negative coupling.

Additional tests are in progress for wings with airfoil fairings to check the qualitative nature of the flat plate results, as well as additional tests with variable sweep to check the combined effects of bending-torsion coupling, stall behavior and sweepback and sweepforward.

## References


1. Krone, N. J., "Divergence Elimination with Advanced Composites," AIAA Paper No. 75-1009, 1975.

2. Weisshaar, T. A., "The Influence of Aeroelasticity on Swept Composite Wings," AFWAL-TR-80-3137, Air Force Wright Aeronautical Laboratories, November 1980.

3. Weisshaar, T. A., "Aeroelastic Tailoring of Forward Swept Composite Wings," Journal of Aircraft, Vol. 18, No. 8, August, 1981.

4. Sherrer, V. C., Hertz, T. J., and Shirk, M. H., "Wind Tunnel Demonstration of Aeroelastic Tailoring Applied to Forward Swept Wings," Journal of Aircraft, Vol. 18, No. 11, November 1981, pp. 976-983.

5. Wilkinson, K., and Rauch, F., "Predicted and Measured Divergence Speeds of an Advanced Composite Forward Swept Wing Model," AFWAL TR-80-3059, Air Force Wright Aeronautical Labortatories, July 1980.

6. Ellis, J. W., Dobbs, S. K., and Miller, G. D., "Structural Design and Wind Tunnel Testing of a Forward Swept Fighter Wing," AFWAL TR-80-3073, Air Force Wright Aeronautical Laboratories, July 1980.

7. Hertz, T. J., Shirk, M. H., Ricketts, R. H., and Weisshaar, T. A., "On the Track of Practical Forward Swept Wings," Astronautics and Aeronautics, Vol. 20, No. 1, January 1982, pp. 40-53.

8. Rainey, G. A., "Preliminary Study of Some Factors which Affect the Stall-Flutter Characteristics of Thin Wings," NACA TN-3622, March, 1956.

9. Dugundji, J., and Aravanudan, K., "Stall Flutter and Nonlinear Divergence of a Two-Dimensional Flat Plate Wing," M.I.T. Aeroelastic and Structures Research Laboratory, ASRL-TR-159-6, July, 1974.

10. Hollowell, S. J., "Aeroelastic Flutter and Divergence of Graphite/Epoxy Cantilevered Plates with Bending-Torsion Stiffness Coupling," M.I.T. Department of Aeronautics and Astronautics, M.S. Thesis, January 1981.

11. Turner, M. D., "Comparison of Static and Dynamic Test Methods for Determining the Stiffness Properties of Graphite/Epoxy Laminates," M.I.T. Department of Aeronautics and Astronautics, M.S. Thesis, June 1979.

12. Tsai, S. W., and Hahn, H. T., Introduction to Composite Materials, Technomic Publishing Co., Westport, CT. 1980.

13. Jensen, D. W., "Natural Vibration of Cantilever Graphite/Epoxy Plates with Bending-Torsion Coupling," M.I.T. Department of Aeronautics and Astronautics, M.S. Thesis, August 1981.

14. Ashton, J. E., and Whitney, J. M., Theory of Laminated Plates, Technomic Publishing Co., Stamford, CT. 1970.

15. Bisplinghoff, R. L., Ashley, H., and Halfman, R. L., Aeroelasticity, Addison-Wesley Publishing Co., Reading, MA. 1955.

16. Crawley, E. F., and Dugundji, J., "Frequency Determination and Non-Dimensionalization for Composite Cantilever Plates," Journal of Sound and Vibration, Vol. 72, No. 1, 8 Sept. 1980, pp. 1-10.

17. Spielberg, I. N., "The Two-Dimensional Incompressible Aerodynamic Coefficients for Oscillatory Changes in Airfoil Camber," Journal of the Aeornautical Sciences, Readers Forum, Vol. 20, No. 6, June 1953, pp. 432-434.

## Acknowledgment

The authors wish to acknowledge the support of the US Air Force and the Materials Laboratory of the Air Force Wright Aeronautical Laboratories, which partially supported this research under Contract no. F33615-77-C-5155.